



Review

Utilizing movement-related potentials to monitor cognitive rehabilitation in individuals with central nervous system disorders: A review

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ABSTRACT

In current neuroscience, there is a pressing need to evaluate the effectiveness of treatments for motor and cognitive disorders. In addition, there is a gap in the literature on assessing this type of rehabilitation. This review proposes using Movement-Related Potentials (MRPs) as a relevant marker for such evaluations. The method employed involves administering a psychomotor task known as the Skilled Performance Task (SPT). Originally described in 1980 by Papakostopoulos, the SPT requires specific abilities, such as bimanual coordination, adaptive programming, and precise timing. Subjects must start and stop a sweep on an oscilloscope screen with accurate timing, receiving immediate feedback that aids in performance improvement. This self-paced, goal-directed, and interactive task ensures continuous engagement and learning. Suitable for all ages, the SPT provides real-time feedback and demands constant attention and reprogramming of movements, making it an effective tool for assessing and promoting cognitive rehabilitation through neuroplasticity. The originality of this task lies in the ability to obtain, with a single motor act, a series of potentials that represent different stages of the processing of a goal-directed and self-initiated action. The MRPs associated with this task are the Bereitschaftspotential (BP), the Motor Cortex Potential (MCP), the N100, the P200, the Skilled Performance Positivity (SPP) and the Post-Action Negativity (PAN). The present study reinterprets previous research conducted on children and adults with various central nervous system pathologies, from a cognitive rehabilitation perspective, to provide new insights into neurorehabilitation techniques utilizing MRPs.

1. Introduction

The last 25 years of the previous century are considered the “golden age” of human psychophysiology. This period was marked by the discovery of Event-Related Potentials (ERPs), such as the Contingent Negative Variation (CNV) (Walter, 1964, 1968) and P300 (Sutton et al., 1965, 1967), to name just the most famous and initially described ones. Subsequently, two approaches have characterized research on ERPs: the physiological and the psychological. The first aimed to study brain responses to variations in the properties of sensory channels, such as receptor and field specificity, adaptation, stimulus threshold, and spatial and temporal resolution. Thanks to their precise temporal resolution, ERPs allowed researchers to measure the exact timing of sensory processing stages, helping to understand how quickly different sensory channels (e.g., vision, hearing) process stimuli and how early sensory processing occurs in the brain. For example, the P100 component in visual ERPs typically occurs around 100 milliseconds after a visual

stimulus, reflecting early visual processing in the occipital cortex.

The psychological approach, on the other hand, focuses on the psychological characteristics of sensory channels—specifically, how sensory information is perceived, processed, and interpreted by the brain, shaping our experience of the world. These properties include the influence of sensory inputs on attention, perception, cognition, emotion, and behavior. Additional psychological characteristics related to each sensory channel include perceptual constancies, habituation, selective attention, neural plasticity, and learning. For instance, the N100 component is linked to auditory processing and is sensitive to levels of attention (Picton et al., 1974; Picton & Hillyard, 1974).

Simultaneously with these studies, another line of research focused on investigating these physiological and psychological properties in individuals with various neurological and psychiatric disorders. The aim was to determine whether different ERP components could represent state or trait conditions and, consequently, whether they could be used as biomarkers (Wacker and Stemmler, 2006). In their comprehensive

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review, the authors discussed how ERP components can be used to study personality traits and the distinction between state and trait influences on ERP measures. Unfortunately, the complexity of these studies and the significant time and financial challenges involved in creating normative databases for each ERP component concerning specific tasks have largely restricted these investigations to comparisons between groups of healthy and pathological subjects. As a result, their diagnostic value has been limited and has been the subject of considerable debate.

However, where ERPs have made a more notable contribution is in the field of psychopharmacology (Iannone and Khanna, 1996; Luck and Kappenman, 2011). The goal was to study the therapeutic effects of psychopharmacological treatments not only through behavioral tests but also through psychophysiological measures. The challenge of using the same subjects as their control was methodologically sound and valuable for assessing changes in various ERP components associated with cognitive processes during a psychopharmacological treatment (Chiarenza and Artoli, 1996; Chiarenza et al., 2016).

While some authors have recognized and suggested the use of ERPs for monitoring and evaluating the effectiveness of neurorehabilitation treatments (Cicerone et al., 2000; Gronholm et al., 2012; Pérez-Serrano et al., 2018; Polich and Kok, 1995), the use of MRPs in this context has been overlooked and only sporadically employed. This article proposes using the Movement-Related Potentials to assess the effects of cognitive rehabilitation in children and adults with central nervous system disorders. Specifically, it introduces a psychomotor task to evaluate various cognitive processes during its execution, along with the associated cortical brain potentials known as Movement-Related Potentials (MRPs). This task, named the Skilled Performance Task, was originally presented in 1980 by Papakostopoulos (Papakostopoulos, 1980b). Additionally, the article presents results from different clinical conditions that demonstrate the suitability of these potentials for measuring the effects of rehabilitation interventions.

This work differs from previous studies by reexamining earlier research through the lens of neurorehabilitation in children and adults with various central nervous system pathologies.

2. Review

To conduct this work, we made an integrative review described in the following Table 1.

The 5 papers found in EBSCOhost were also found by the Europe PMC, which found 2 additional ones. The seven papers have been referenced in the present work.

Before starting cognitive rehabilitation, a thorough assessment of the

Table 1
Indicates the databases, the descriptors, and the number of papers retrieved for the traditional review.

Databases	Search strategy	Selection criteria	Number of papers retrieved
EBSCOhost Research Platform ¹	((“movement-related potentials” OR “movement potentials”) AND “skilled performance task”))	ALL	5
Europe PMC			7

¹ EBSCOhost Research Platform includes: Academic Search Premier, Anthropology Plus, APA PsycArticles, APA PsycInfo, Business Source Ultimate, CINAHL Complete, eBook Academic Collection (EBSCOhost), eBook Collection (EBSCOhost), eBook Open Access (OA) Collection (EBSCOhost), EBSCO eClassics Collection (EBSCOhost), Education Source, ERIC, European Views of the Americas: 1493 to 1750, GreenFILE, Library, Information Science & Technology Abstracts, MEDLINE, MLA Directory of Periodicals, MLA International Bibliography with Full Text, OpenDissertations, Political Science Complete, PSICODOC, Psychology and Behavioural Sciences Collection, Regional Business News, PRISM Series A/II: Music Manuscripts after 1600, SPORTDiscus with Full Text, Teacher Reference Center, The Serials Directory, and the Education Source Ultimate.

patient’s sensory and cognitive functions is essential. Typically, neuropsychological test batteries examine various functions, such as motor, perceptual, mnemonic, attention, and linguistic. However, using neurophysiological tests is less common due to the need for specialized laboratories, expensive equipment, and expertise in anatomy, physiology, and neurophysiology.

The sensory and cognitive assessment involves a detailed analysis of impaired and residual functions using neurophysiological tests that have been experimentally validated on the patient. To define an effective rehabilitation strategy, it is crucial to identify the impaired cognitive processes at various stages of information processing and the residual capacities that can be leveraged. Cognitive rehabilitation should engage the patient’s active participation by utilizing all available residual capacities.

There are two main approaches to formulating a rehabilitation plan: one aims to recover the impaired function, based on modular theory (Casanova and Casanova, 2019), while the other activates and stimulates unimpaired cognitive functions to promote compensation (Jenkins and Merzenich, 1987). Regardless of the approach, a cognitive rehabilitation project must include active patient participation to maintain motivation and interest in performance improvement. Cognitive rehabilitation is based on the notion of neuroplasticity (Nahum et al., 2013).

Neuroplasticity refers to the brain’s capacity to alter its structure and function in response to internal and external stimuli, such as learning, experience, or recovery from injury. This phenomenon enables changes in synaptic strength, the formation of new neural pathways, and even structural changes within brain regions. Studies have shown that plasticity can improve cognitive function and motor abilities following events like stroke or brain trauma, with neural adaptation occurring in areas that remain healthy to compensate for damaged regions (Doide, 2007; Kolb and Gibb, 2011). Initially, sensory maps were thought to be immutable, but recent studies in humans and animals have shown that the somatosensory, auditory, and visual cortices can remodel following peripheral or central lesions or congenital defects (Buonomano and Merzenich, 1998; Neville et al., 1983; Röder et al., 1999). This neuroplasticity enables the reprogramming of specific brain functions and the establishment of permanent changes in cortical neuron connections, leading to behavioral changes in learning.

Another external factor contributing to neuroplasticity may be nutritional modulation and metabolic adaptation to nutritional factors may be an internal response related to the stimulus. A recent study in rodents suggests that high-dose riboflavin stimulates sensorimotor and somatic development (da Silva-Araújo et al., 2024).

Nahum and Merzenich highlight the prerequisites for an effective rehabilitation plan: attention, accuracy, adequacy, reinforcement, frequency, and flexibility (Nahum et al., 2013). Additionally, a successful rehabilitation program must be interactive and intentional to stimulate motivation and performance improvement. Cognitive rehabilitation is grounded in the latest notions of neuroplasticity.

3. The Skilled Performance Task

We will describe the Skilled Performance Task (SPT) (Papakostopoulos, 1980a) and the associated movement-related potentials. This task, due to its characteristics, appears well-suited for evaluating the effectiveness of a rehabilitation program. The SPT is self-paced, voluntary, goal-directed, and interactive. To perform it successfully, the task requires the following skills: bimanual coordination, bimanual ballistic movements, adaptive programming, learning proper timing, and performance improvement. The task provides online knowledge of results and feedback. It involves starting a sweep of the oscilloscope trace by pressing a button with the left thumb and stopping it by pressing another button with the right thumb. This must be done within a target area on the oscilloscope screen, 40–60 ms from the onset of the sweep. The sweep velocity is 1 mm/ms. Fig. 1 illustrates the SPT setting for recording the MRPs.



Fig. 1. SPT setting: The subject sits comfortably in front of the oscilloscope (1), with a mechanical pressing button in each hand (2 and 3). The EEG is recorded (4) during task execution. The 2 yellow bars in the middle of the oscilloscope screen (b) indicate the target interval. The areas marked with “a” and “c” indicate the wrong interval, either too fast (a) or too slow (c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Since the task is self-initiated and voluntary, it demands constant attention to perform it correctly. The rapid sweep speed (1 mm/ms) compels the subject to learn the precise interval between the two button presses using all available feedback, including kinaesthetic and visual information from thumb presses and from stopping the sweep on the target area on the screen, respectively. Therefore, accuracy is achieved by learning this timing interval and measuring the number of successes within the target area and the number of performances around it. Reinforcement comes from real-time visual feedback as the sweep stops on the screen, requiring the subject to reprogram movements based on whether the result was too fast (0–39 ms) or too slow (61–100 ms).

This knowledge of the results introduces considerable flexibility in the subjects, as they must adjust the two movements based on the outcome: either too fast (when the sweep is stopped before the target area, i.e., 0–39 ms) or too slow (when it is stopped after the target area, i.e., 61–100 ms). Given the sweep speed (1 mm/ms), the subject must intentionally pre-program both the start and stop movements as a single, continuous movement. The programming, organization of movement, and flexibility represent one of the most complex executive functions of the frontal lobes. These characteristics define the task as interactive and motivate the subject to engage in frequent training to improve performance, thus preventing habituation or, even worse, boredom.

Finally, the task should be appropriate for individuals of all ages, as excessive difficulty may impede learning and, consequently, limit changes in neural maps. The task is designed for participants ranging from 6 years old to older adults, with its difficulty calibrated to prevent the occurrence of a “ceiling effect.”

This task activates sensorimotor and cognitive functions which are deeply intertwined, suggesting that thinking and perceiving are not just connected to action but fundamentally rely on it. The connection between these functions stems from how the brain develops, processes information, and organizes experiences.

The relationship between motor and cognitive functions in the brain is evident in conditions like Parkinson’s disease (PD), which demonstrates how closely intertwined these systems are. PD impairs not only motor abilities, such as movement and coordination, but also cognitive functions, including attention, memory, and planning. (Kalia and Lang, 2015). Research indicates that disruptions in neurotransmitter systems, such as dopamine and noradrenaline, are linked to both motor and cognitive deficits in PD, suggesting that brain structures involved in motor control also contribute to cognitive processes (Kehagia et al.,

2010). This interdependence emphasizes that cognition is fundamentally rooted in sensory and motor interactions, supporting the notion that thinking and perceiving are grounded in action (Kalia and Lang, 2015; Kehagia et al., 2010). Additionally, studies on individuals with sensory impairments, such as visual or auditory deficits, reveal that the brain often adapts by reorganizing sensory processing to maintain cognitive functions. This adaptive plasticity highlights the brain’s capacity to use remaining sensory inputs to support cognition, underscoring the role of sensorimotor integration in shaping cognitive abilities (Bavelier and Neville, 2002; Collignon and Zatorre, 2015). This phenomenon aligns with the theory of embodied cognition, which posits that cognitive processes are not isolated but deeply rooted in physical and sensory experiences (Wilson, 2002; Barsalou, 2008).

In conclusion, there is evidence supporting the claim that motor and cognitive functions are interconnected and that both sensorimotor and sensory experiences play crucial roles in cognitive processing and adaptation. This body of research also underpins therapeutic approaches that aim to improve cognitive functions by targeting motor and sensory processing.

4. Movement-Related Potentials Components

Overall, the SPT engages various cognitive processes that correspond to specific brain potentials and defined Movement-Related Potentials, making it an effective tool for assessing rehabilitation outcomes. The appendix describes the SPT and the recording of Movement-Related Potentials in detail.

Based on the spatial-temporal characteristic of these potentials and their relationship to electromyographic activity and performance, four successive periods can be identified during a skilled task: the premotor period, the sensory-motor period, the motor completion period, and the post-motor period (Fig. 2) (Papakostopoulos, 1978).

5. The premotor period

The Bereitschaftspotential (BP) is characteristic of the premotor period, with its onset preceding EMG activities of 800–1200 ms (Kornhuber and Deecke, 1965). The BP is present only during active, self-paced, voluntary movements and is absent during passive movements (Papakostopoulos, 1980a). Its amplitude is higher during skilled and goal-oriented tasks than during unskilled and non-purposeful ones (Chiarenza et al., 1980). The BP’s scalp distribution is prevalent in the central and precentral areas. Recent studies indicate that BP could consist of at least two components. The first begins 1.2 s before the movement and lasts for about 450 to 600 ms while the second is characterized by a steeper negative ramp lasting 300–400 ms. The first component is associated with processes related to the representation of the action and motor programming, while the second is considered the most automatic and effective part of the process (Shibasaki, 2012). Intracerebral recordings of the BP have confirmed this observation (Kukleta et al., 2012, p. 20). Recently, Paradiso et al. (Paradiso et al., 2004) have demonstrated the involvement of the thalamus in the preparation of self-paced movements. Neither the scalp nor the thalamus showed pre-movement potentials with passive wrist extensions. The thalamic MRP preceded both contralateral and ipsilateral voluntary wrist movements. There was no significant difference between the onset time of thalamic MRP and cortical MRP, with maximum amplitude occurring at contacts located in the ventral lateral nucleus. During unimanual action, the BP is predominantly in the contralateral hemisphere. However, during bimanual skilled action, the BP is symmetric in the right-hand subjects but not in the left-hand subjects (Chiarenza et al., 1980, p. 198; Papakostopoulos, 1978). It has been proposed that this potential seems to reflect the intention to act (Libet, 1983), the ability to program, i.e. the strategic organization of the ideokinetic elements necessary to achieve the goal (Chiarenza, 1998, 2022; Chiarenza et al., 1995). The BP is age-dependent and modifies with practice (Chiarenza,

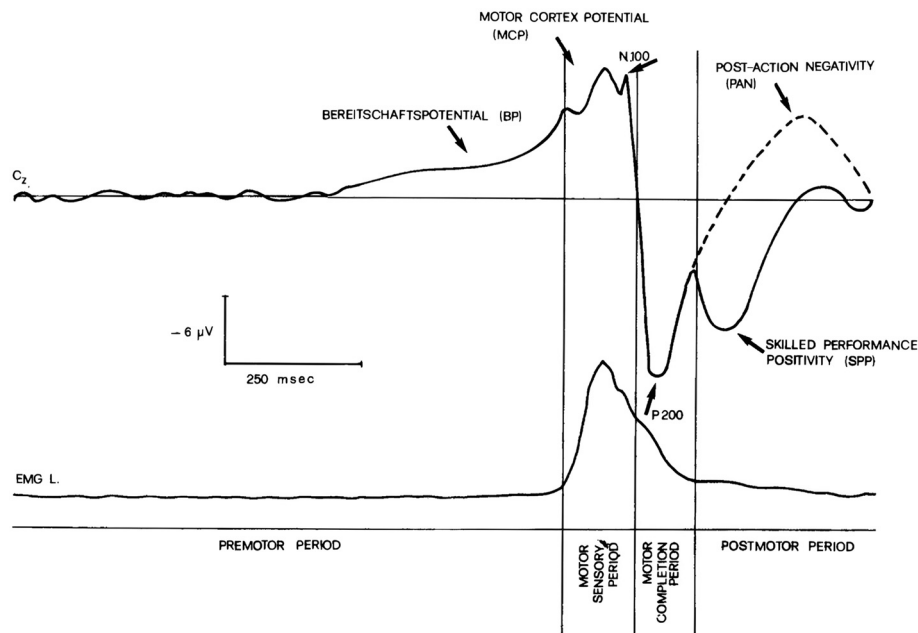


Fig. 2. Schematic diagram of Movement-Related Potentials related to the Skilled Performance Task.

1998). In children aged six to seven, the BP is barely visible (Chiarenza et al., 1995). Therefore, despite the awareness of movement in these children, the BP characteristic of the preparatory period is almost absent (Chiarenza et al., 1995). Around 9–10 years of age, BP starts to emerge in the frontal and central areas. During this age, children acquire the adult capacity of abstract thinking (Bruner, 1970). They conceive many possible ways in which they could operate and many alternative ways in which they could perform better. We suppose that these different possibilities, these “strategic qualities,” i.e., the intentionality of action to achieve a goal, are reflected in the BP amplitude. Performance is known to improve with practice and distributed practice is more efficient than intensive practice for learning (Chiarenza, 1998). Practice significantly modifies the amplitude of BP. In a study with nine subjects, aged 11–14, participants completed 100 trials of SPT, repeating them after 1 h, 14 days, and again after one month. The succession of 100 trials was considered massive practice, while the other trials were considered distributed practice. With both types of practice accuracy and number of target performances improved. The effects of practice on the BP were observed only after one month. The BP shows a smaller amplitude and reduced onset in all cerebral regions (Chiarenza, 1998). These results seem to indicate that a certain time is necessary for “the motor engram” related to the task to be memorized; afterward, the organization of a skilled performance requires less time, reduced BP onset, a smaller effort, and reduced BP amplitude. It is known that mnemonic trace storing does not happen all at once but occurs gradually after the initial learning phase (Sumner et al., 2020). This phenomenon of consolidation is attributed to reverberating circuits, which increase the efficiency of certain synapses, causing permanent structural changes in cerebral circuits. These changes seem to represent the neuronal basis of long-term memory (Lynch, 2004).

These observations have been corroborated by Fattapposta et al. (Fattapposta et al., 1996) who observed that Olympic gun-shooting champions, during long-term practice with SPT, achieved a higher number of target performances compared to the control group. These elevated performance levels were associated with a decrease in BP amplitude. They interpreted these findings to indicate that Olympic champions with extensive practice recruited fewer resources to achieve optimal performance, as reflected in the BP magnitude.

Despite the significant potential of this task for the electrophysiological study of executive functions, the SPT has not been widely utilized

to validate these results in normal subjects and to demonstrate its utility in the clinical field as a measure for therapeutic and rehabilitative treatments.

In adults, BP has been used to monitor the effects of motor rehabilitation in hemiparetic patients post-stroke (Fattapposta et al., 2008), both before therapy, and at 3, 9, and 12 months following therapy. The results highlighted a characteristic trend for BP during rehabilitation. At baseline, hemiparetic patients scored lower in target performances and had a lower BP amplitude compared to controls; the number of correct performances did not significantly improve during subsequent recordings. BP amplitude mildly increased in the second, third, and fourth recordings. In these hemiparetic subjects, the ability to plan a movement seems much more limited than in controls.

This method has been applied also to psychiatric disorders such as chronic and early schizophrenics (Chiarenza et al., 1985; Chiarenza and Cazzullo, 1989). In subjects with chronic schizophrenia the BP was absent or significantly reduced in all brain areas, while in subjects after the first episode of schizophrenia, the BP was reduced only in left precentral areas.

In children, this task has been employed to study children with developmental dyslexia (Chiarenza, 1990; Chiarenza et al., 1982a, 1982b). The results indicate that children with developmental dyslexia exhibit a lack of programming, as evidenced by a significantly reduced BP amplitude and onset. The BP is present in its automatic component, approximately 200 ms before the onset of electromyographic activities. These findings were later confirmed by Nicolson and Fawcett (Nicolson and Fawcett, 2005) who concluded that dyslexic children face difficulties when required to undertake fast, fluent, over-learned skills, or novel skills that involve the blending of two actions. Their performance after extensive practice remains slower and more error-prone, indicating lower skill proficiency and the ‘quality’ of automatized performance. It seems reasonable to argue that dyslexic children have difficulties both with the initial proceduralisation of skill and with the ‘quality’ of skill post-training.

6. The sensory-motor period

The premotor period is followed by the sensory-motor period starting from the onset of phasic electromyographic activity and lasting for about 200 ms. It is during this period that behavior becomes manifest. It

coincides with the appearance on the scalp of the Motor Cortex Potential (MCP) and N100. The MCP, a negative potential follows the BP and peaks 40–80 ms after the EMG peak. It is present during unskilled actions and increases in amplitude during self-paced, ballistic, and sustained motor performance (Chiarenza et al., 1980; Grunewald-Zuberbier et al., 1979). It is observed in both children and adults, with its amplitude decreasing with senescence (Chiarenza et al., 1980). Scalp, cortical, and magnetic recordings of MRPs with the same skilled task have clearly shown that the source of the MCP is located anteriorly to the central fissure (Chiarenza et al., 1991; Papakostopoulos and Crow, 1984; Rektor et al., 1998). This potential seems to be an index of reafferent peripheral input from the skin, tendons, and muscle receptors (Papakostopoulos and Crow, 1984).

In our opinion, this potential could serve as an interesting index for evaluating the efficiency of rehabilitation plans and testing sensory-motor integration processes. The results of movement-related potentials in subjects with Down syndrome are noteworthy in this context (Chiarenza, 1993). Comparing the MRPs of subjects with Down syndrome with those of two control groups, one matched for mental age and the other for chronological age, revealed differences attributable to Down Syndrome rather than chronological or mental age differences. The performance of the SPT presupposes that the subjects understand the requirements to accomplish the task and can maintain their commitment to the task. Subjects with Down syndrome performed the SPT task according to instructions but required significantly more practice time. Successful performance depends largely on the correct sequence of two ballistic movements, which requires fine-tuning of a central clock to control information from afferent and efferent systems, i.e., the information from the two finger presses (Hirsch and Sherrick Jr., 1961; Rosenbaum and Patashnik, 1980). The interval between these movements is too short (40–60 ms for target performance) for feedback from the first movement to trigger the second, necessitating the programming of both movements as a single unit (Chiarenza, 1993; Chiarenza et al., 1991; Kelso et al., 1979). Throughout the task, subjects with Down syndrome exhibited significantly lower accuracy and speed than both control groups, with performance not improving with practice. Most performances fell within the 0–20 ms or > 100 ms intervals, resulting in a low number of target performances (Chiarenza, 1993). This suggests that individuals with Down syndrome struggle to tune the central clock to control information from the two movements. Moreover, this study confirms that Down syndrome subjects have difficulties when programming a sequence of movements to coincide with an external event (Henderson et al., 1981a). The specific motor programming issue in Down syndrome appears to be temporal rather than spatial (Henderson et al., 1981b). Another explanation for their low performance is the observation of MRPs. Subjects with Down syndrome exhibited an absent or greatly reduced BP and the MCP an index of proprioceptive afferent information processing was absent in the precentral cortex. Adequate proprioceptive information is crucial for the preparation and execution of movements. The absence of MCP indicates poor temporal organization of complex motor acts. Animal studies and observations of patients with posterior spinal cord column damage show that deafferentation impedes the temporal control of motor sequences (Dubrovsky and Garcia-Hill, 1973). The absence or reduced MCP amplitude in Down Syndrome subjects is also linked to the anatomopathological condition of their cerebral cortex, characterized by a reduced number of synapses, and early Alzheimer's phenomena (Becker et al., 1986). The MCP is the first potential to decrease in amplitude in adults over 60 years (Papakostopoulos and Banerji, 1980).

MCP is followed by the cortical evoked potential, N100, the expression of the early stages of visual perceptual processing of the brain response to the visual stimulus, i.e., the oscilloscope's sweep. This potential, due to the inhibition phenomenon, is reduced in amplitude in the precentral areas (Papakostopoulos et al., 1975). The quality and

intensity of inhibitory phenomena on the motor cortex during movement are also parameters that should be further investigated during a rehabilitation treatment. They are a further expression of sensory-motor integration mechanisms. For example, in children with developmental dyslexia, also this process seems impaired (Chiarenza, 1990; Chiarenza et al., 1982b). During the sensory-motor period, the latency of the visual evoked potential to the oscilloscope's sweep is significantly increased in prefrontal, frontal, central, and precentral areas compared to the control group. Latency expresses the speed with which information is carried out through the nervous system. The visual response in dyslexic children reached different cortical areas at different times. The reduced amplitude and the increased latency of N100 together with a reduced amplitude of the Motor Cortex Potential could indicate a defect in the integration of reafferent kinaesthetic information.

7. The motor completion period

The motor completion period is characterized by the decline of the EMG activities and a positive peak with a latency of 200 ms (P200). This potential seems to reflect the reafferent activity of the deeper structures of the Central Nervous System (Vaughan Jr. et al., 1968). In children with developmental dyslexia, the latency of P200 in the prefrontal areas is significantly increased and its amplitude is reduced across all other recorded cerebral areas. The presence of P200 in the absence of external stimuli (Vaughan Jr. et al., 1968) and its increased amplitude with force (Wilke and Lansing, 1973) supports the hypothesis that P200 is a reafferent somatosensory potential related to movement. The reduced amplitude in dyslexic children across various brain areas, even during the same performance interval (target performance), could indicate a defect in the integration of reafferent information. The ability to process and integrate somatosensory information is a prerequisite for any rehabilitation plan.

8. The post-motor period

The post-motor period is characterized by the return of the electromyographic activities to the preceding rest conditions and a large positive potential with a latency of 460 ms, termed Skilled Performance Positivity (SPP) (Papakostopoulos, 1980a). This potential has a central and parietal scalp distribution. Magnetic signals, recorded over the hand somatomotor area during the SPT, indicate that SPP originates from the somatomotor cortex (Chiarenza et al., 1991). This indicates that motor, cognitive, and emotional functions of the brain are performed in an integrated manner in the somatomotor cortex. Therefore, classical differentiation between sensory, motor, and cognitive activities seems inadequate in this context.

SPP is observed in healthy subjects when they are asked to perform a motor-perceptual task that requires accuracy, timing, and performance level improvement by providing adequate real-time feedback information on the outcome. Because of these properties, SPP, together with BP, can represent a psychophysiological index sensitive to behavioural changes due to rehabilitation treatment. The SPP is absent during unskilled tasks and when adequate feedback is not provided to the subject (Papakostopoulos, 1980a; Papakostopoulos et al., 1986). From a chronological standpoint, SPP coincides with the subject's awareness of the success or failure of his performance. This positivity appears when the subject looks for information about his outcome, i.e. information relevant to the efficiency of his psychomotor pre-programmed organization (Chiarenza et al., 1990). Knowledge of the outcome is likely used to influence future actions. Up to 9 years of age, SPP is present only in the posterior areas, with its amplitude increasing with age while its latency decreases significantly. Around 10 years of age, SPP also appears in the prefrontal and frontal regions likely related to the maturation of long-distance connections between the parietal and frontal areas. Parietal

areas primarily handle associative somatosensory and visual perceptual functions, while intention, motivation, attention, programming, and evaluation of voluntary and goal-directed behavior are functions of the frontal lobes (Fuster, 1985, p. 1). These areas have a different maturational trend: parietal areas reach adult levels of synaptic formation around 10–12 months of age, while frontal areas do so at around 10 years of age (Thompson and Nelson, 2001). Therefore, in parietal areas, SPP may express more perceptual functions, while in the frontal regions, it may represent more abstract functions. This viewpoint suggests that the emergence of SPP in frontal areas corresponds to a new brain functioning mode: using outcomes and knowledge of results to improve goal-directed behavior by matching them with past experiences and projects. (Chiarenza, 2022; Chiarenza et al., 1990). After 11 years of age, thinking becomes propositional. During the period of formal thinking, children consider the outcome of their performance, form sentences, and begin finding relationships among them, thus using experience for learning. (Inhelder and Piaget, 1958).

This interpretation is supported when the behavior of MRPs related to performance accuracy in children and adults was compared i.e. comparing the trend of SPP during target performance and near-target performance (Chiarenza et al., 1990). In children, SPP amplitude was significantly greater during the target performance in frontal and right precentral areas and decreased with increasing inaccuracy of performance. In central and parietal areas, there was no significant relationship between SPP amplitude and performance time. In adults, no significant systematic relationship was observed between SPP amplitude and performance time across various brain areas. SPP latency related to performance showed a different trend in adults and children. Except in prefrontal and frontal areas, polynomial regression indicates that SPP latency had an upward parabolic trend in adults and a downward one in children, peaking during target performance in both groups. This parabolic trend shows that SPP peaks in children during target performances, while in adults, it peaks during incorrect performances. The difference in SPP latencies between adults and children is not due to the number of ‘wins’ as both groups have nearly the same number of target performance events. SPP latency seems to depend on the time needed to evaluate the information conveyed by the stimulus, with adults deliberating on the reasons for errors and considering strategy changes. This behavior is not seen in children, who prioritize successful events and immediately proceed to the next attempt. The emphasis children place on winning is reflected in SPP amplitude, which is highest during target performances and decreases with an increasing range of errors. In adults, no relationship was found between performance quality and SPP amplitude; it remained consistent across all brain areas during both target and incorrect performances, indicating equal relevance given to both outcomes. The relevance of that study in the context of cognitive rehabilitation lies in showing how accuracy can be measurable in milliseconds and how variations in SPP amplitude and latency can elucidate cognitive processes related to performance evaluation.

Another performance-related potential with negative polarity, present in children aged 6–9 during the post-motor period, in the prefrontal and frontal regions is Post-Action Negativity (PAN) (Chiarenza et al., 1984). This negative potential is linked to poor performance and has been associated with rare, novel, unexpected events or error detection (Courchesne et al., 1987; Dehaene, 2018; Hillyard and Kutas, 1983). PAN may represent the pre-conceptual stage of representational intelligence, in which reality depends on immediate perception (Inhelder and Piaget, 1958). The surprise or novelty of the outcome of the performance predominates in these children, with PAN decreasing with age and disappearing by 8–9 years.

In summary, the BP seems to reflect the intention to act or the preparation for movement. The MCP is an index of reafferent peripheral input from the skin, tendons, and muscle receptors. The N100 is the brain response to the visual stimulus while P200 seems to reflect the reafferent activity of deeper structures of the Central Nervous System. The SPP is related to knowledge and evaluation of the performance

Table 2

Summarizes the electrophysiological characteristics of the MRPs.

Movement-Related Potentials: chronogram

- **Premotor Period**
 - Bereitschaftspotential (BP)
 - BP precedes voluntary movements
 - Reflects Intention to Act: Preparation for voluntary movements
 - Age-dependent and modifiable with practice: Emerges around 9–10 years old
 - Modifiable with Practice: Intensive practice leads to reduced BP amplitude
 - Types of Practice
 - Massive Practice: High frequency over a short period
 - Distributed Practice: Spaced intervals over a longer period
 - Practice and Learning: BP and Performance: Lower BP amplitude with skilled performance
 - Memory Consolidation
 - Mnestic trace stored through reverberating circuits
 - Neural changes over time with repeated practice
 - Thalamic MRP observed
- **Sensory-Motor Period**
 - Motor Cortex Potential (MCP): Generated during motor action
 - N100 Potential: Reflects sensory processing
 - Differences in Dyslexic Children: Altered MCP and N100 responses
- **Motor Completion Period**
 - P200 Potential: Reafferent sensory activity post-movement
 - Indicates completion of motor act
- **Post-Motor Period**
 - Skilled Performance Positivity (SPP): Reflects performance monitoring and adjustment, Linked to performance evaluation processes
 - Age-Related Changes of SPP: Developmental trajectory of performance monitoring
 - Post-Action Negativity (PAN): observed in younger children and in children with poor performance

Studies and Observations

- **Olympic Gun-Shooting Champions**
 - Low BP amplitude linked to high skill level
 - Efficient neural resource utilization
- **Hemiparetic Patients Post-Stroke**
 - Rehabilitation leads to BP changes
 - Gradual increase in BP amplitude with recovery
- **Children with Developmental Dyslexia**
 - Reduced BP amplitude and delayed onset
 - Indicates motor programming difficulties

Rehabilitation and Cognitive Processes

- **BP, MCP, N100, P200, SPP, PAN as Indices**
 - Indicators of motor and cognitive recovery
 - Useful in assessing rehabilitation progress and efficacy

Potential Applications

- **Therapeutic Use of MRPs:** For motor and cognitive rehabilitation
- **Motor Skill Training:** Use in optimizing practice schedules
- **Diagnostic Tool:** Assessing psycho-motor function in neurodevelopmental disorders and cognitive impairment in adults.

outcomes and the PAN is associated with unexpected events or error detection. A more detailed description can be found in Table 2.

The normal performance of this task elicits the Bereitschaftspotential (BP), or premotor potential before the onset of movement and the Skilled Performance Positivity (SPP) after movement, which coincides with the subject's awareness of the success or failure of his performance. The non-presence or abnormalities in these potentials are highly associated to cognitive problems (Chiarenza, 2022; Chiarenza, 2023; Paradiso et al., 2004).

9. SPT and MRPs methodological features

The SPT is self-paced and voluntary. All the other experimental designs, both neurophysiological and behavioural, are externally-paced i.e. built on the stimulus-response model. They can describe only

phenomena that occur in the interval between the stimulus, visual or auditory, and the response of the subject, without being able to observe the phenomena before the onset of the stimulus and after the onset of the response. In this way, only phenomena related to the processing of auditory and visual stimuli can be described.

SPT and MRPs provide neurological insights that surpass cognitive test scores by capturing brain activation and connectivity changes directly related to motor-cognitive functions in rehabilitation. MRPs, for instance, offer valuable data on neural dynamics in motor areas, clarifying cortical changes during rehabilitation, especially when motor impairments impact cognitive performance (Di Russo et al., 2017). They pinpoint shifts in premotor and motor cortices resulting from therapeutic interventions. In contrast, cognitive tests like the Stroop Test (Stroop, 1935) or Wisconsin Card Sorting Test (Berg, 1948; Heaton et al., 1993) provide behavioral insights but don't directly assess motor-cognitive mechanisms. As indirect measures, they may miss subtle neural adaptations, limiting their value in real-time rehabilitation adjustments (Chang et al., 2021).

SPT and MRPs enable clinicians to monitor motor-cognitive recovery effectively, aligning closely with rehabilitation goals. SPT's task-based performance metrics and MRP's neural monitoring allow for detailed tracking of recovery, supporting personalized therapy adjustments (Wu et al., 2021). Cognitive tests alone may not capture these integrated improvements, particularly when motor recovery is prioritized. Tools like the Montreal Cognitive Assessment (MoCA) (Nasreddine et al., 2005) or Mini-Mental State Examination (MMSE) (Folstein et al., 1975)

offer limited insight into motor progress, making them less effective for real-time tracking in conditions like stroke or TBI (Chan et al., 2008; Rinne et al., 2011).

10. Neurorehabilitation and its interactions with the motor and cognitive processes of the brain

As a summarizing view of all aspects discussed in this work, we present a graphical diagram of neurorehabilitation to illustrate the relationships and flow between intention, attention, frequency, accuracy, flexibility, feedback, and adequacy, emphasizing their collective contribution to an effective neurorehabilitation strategy. Each element is interconnected, highlighting the comprehensive and multi-faceted approach required for optimal outcomes. (Fig. 3).

11. A new horizon for the Skilled Performance Task and the Movement-Related Potentials

Recently a neurorehabilitation program called Kinesthesia has been proposed for the rehabilitation of motor and cognitive functions of the brain (Alvarez et al., 2014; Álvarez et al., 2013; Guerrero Sauzamedia et al., 2017, 2018; Navarrete-Torres et al., 2018). This dynamic and interactive system consists of video games based on kinematics, using motion sensors to determine the position of the player (patient) in space. The program's uniqueness and originality lie in the understanding that motor and cognitive activities in the brain are not isolated but

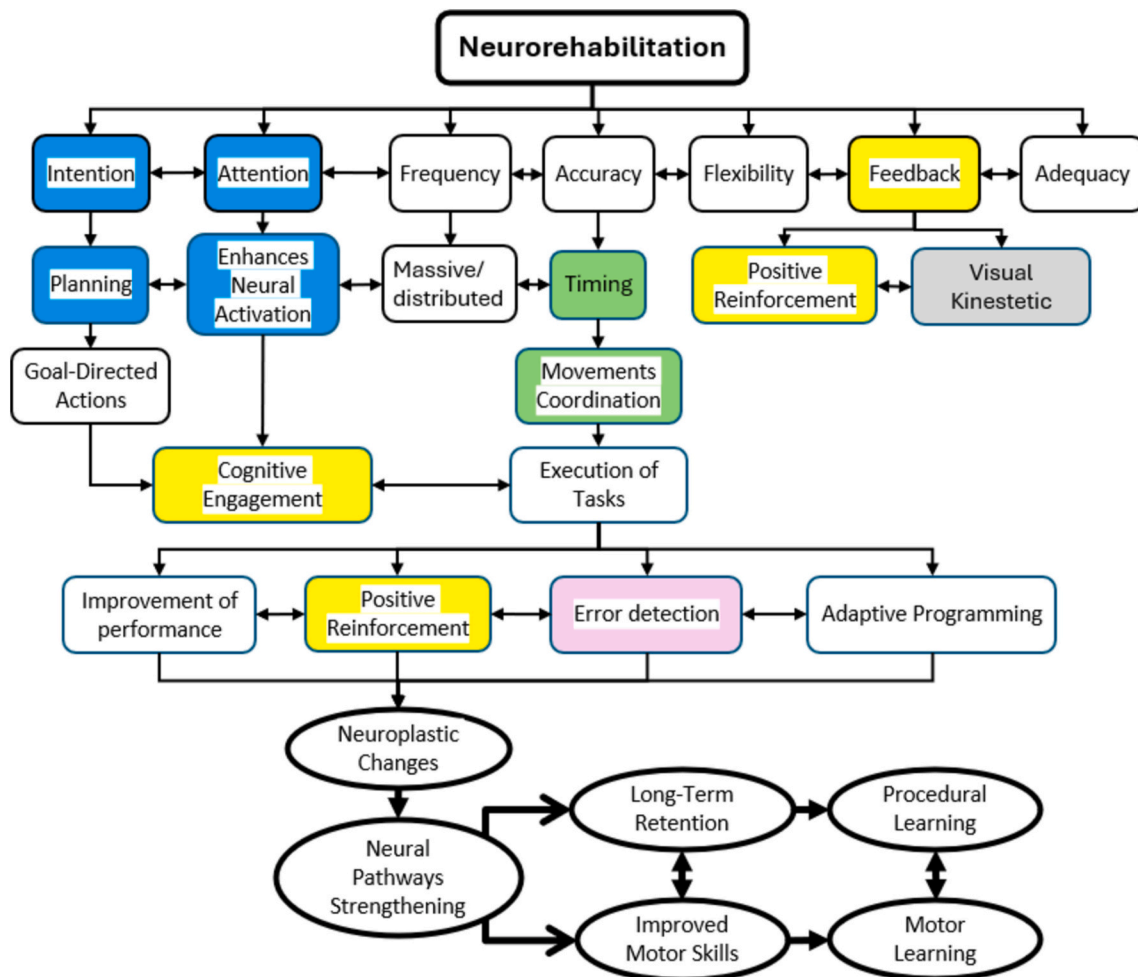


Fig. 3. Flow and relationship of cognitive processes involved in executive functions considered in a neurorehabilitation program. The colors in the boxes refer to the stage of MRPs that can be measured: Blue = BP, green = MCP, grey = N100 and P200, yellow = SPP, pink = PAN. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

integrated. Cognitive activities involve several cerebral areas for observation, planning, and execution, activating not only motor networks, including the cerebellum and basal nuclei but also frontal lobes for execution planning and task-irrelevant stimuli inhibition, thereby improving attention. Motor and cognitive tasks are presented simultaneously, with real-time feedback provided, requiring performance improvement from the subject.

This program, with a solid neurophysiological basis, has the requirements for complete cognitive rehabilitation, considering the multiplicity of cognitive processes involved in movement. The SPT can be considered one of the most suitable tools for evaluating the effectiveness of the Kinesthesia program.

12. Limitations and recommendations

We did not perform a systematic review according to PRISMA guidelines (Moher et al., 2009). However, we did exhaustive research on the international databases described in the methodology. Very few papers have been published on this topic and nearly all the consulted databases reported the same papers.

The hardware used to measure the MRPs was developed in our laboratory and is not easily available in the market. However, with the information provided in this paper, it would be possible to reproduce this setting in other laboratories.

CRediT authorship contribution statement

Giuseppe A. Chiarenza: Writing – review & editing, Writing – original draft, Conceptualization. **Jorge Bosch-Bayard:** Writing – review & editing.

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Declaration of competing interest

The author declares no conflict of interest.

Appendix A

The skilled performance task

The subject sat in an armchair placed 70 cm from an oscilloscope in a lighted and electrically shielded room, holding a push-button in each hand. The excursion of the button was 5 mm. The task consisted of starting a sweep of the oscilloscope trace by pressing the left-hand button with the left thumb and stopping it by pressing the right-hand button with the right thumb, within a predetermined area of the oscilloscope screen, 40–60 ms from the onset of the sweep; the sweep velocity was 1 mm/ms. Performance within this time interval was defined as the “target performance.”

After a verbal explanation and practical demonstration of the task, subjects were given enough time to master the task to achieve a sufficient number of target performances. None of the subjects had previous experience with this task or any other type of motor test. The practice also enabled the subjects to learn how to control eye movements and blinking while executing the task and to pace their trials at intervals of between 7 and 20 s. Subjects were asked to remain relaxed during the task and avoid muscular preparatory movements before pressing. The task ended when 100 trials artifact-free were collected. Four blocks (set) of 25 trials each were averaged and analysed.

EEG recording

Ag/AgCl electrodes were fixed to the scalp with collodion over the

prefrontal (Fpz), frontal (Fz), central (Cz), right precentral (RPC), left precentral (LPC), parietal (Pz), right parietal (P4), and left parietal (P3) regions, according to the International 10/20 system. Each electrode was referred to the linked mastoid. A bipolar EOG was also recorded. The EMG was recorded from the left and right bodies of the long flexor muscle of the thumb. Electrode impedance was <3 k Ω . The EEG was amplified with Physio-Amp Marazza preamplifiers: high-pass and low-pass filters (6 dB/octave) were respectively 0.019 Hz–70 Hz for the EEG and 5.3 Hz–700 Hz for the EMG. The analysis started with sampling for each channel, a square wave of ± 25 μ V for calibration, and equalization procedures. A trigger pulse generated by the left-hand button press was used to initiate the data samples for each channel. The sampling rate was 250 Hz for 2.2 s preceding the trigger pulse and 1 s immediately following it. An average of the first 1 s of the sample was used to establish a baseline from which all amplitudes were measured. Amplitude values were equalized across channels based on a stored calibration pulse.

Data analysis

Performance was measured as the time interval between the two presses and defined as “performance time” (PT). Target performance (TP) was defined as the number of tries in which the sweep was terminated within the correct 40–60 ms target interval. Sweep terminations <40 ms or >60 ms were designated as errors. Accuracy, defined as “performance shift” (PS), was calculated as the distance of the end of the sweep from the target area. 100 consecutive trials without muscular artifacts, blinking, or eye movements were averaged and measured for each subject. The amplitude and onset of BP were measured as follows: the moving average method was employed to determine BP onset. It was established that BP onset, if present, should appear within a time frame of 1200 ms before EMG onset. Two averages were calculated for each point in this range, the first being the mean of the potentials at all points preceding the fixed one, excluding the last nine, and the second being the mean of the potentials at the 20 points around the fixed point (nine preceding the fixed point itself and ten following). Consequently, the point at which the second mean constantly remained greater as an absolute value than the first was chosen as that corresponding to the onset of BP. The twenty-point value corresponding to an interval of 80 ms was chosen based on experimental data because the result was ideal for correctly identifying the onset of BP. To isolate the left EMG onset, the same moving average method was used. In each step, the slope of a simple linear regression was calculated. A significant change in the slope indicated EMG onset. The mean BP amplitude was computed for 200-ms periods immediately preceding the left EMG onset.

In this motor-perceptual task, the temporal-sequential organization of movements of both hands or bimanual coordination, has, for various reasons, a fundamental importance for the correct execution of the task. The two movements are not separated, but integrated, ballistic, and thus preprogrammed. Based on these considerations, the corresponding cerebral electro-activity is understood to be a global expression of the executed task. Therefore, the measurements of the various potentials have been made keeping in mind the chronological evolution of the performed task. Thus, the mean amplitudes were computed for 200 msec periods, immediately preceding the left EMG onset for the BP and immediately following it for the MCP. P200 amplitude was measured as the difference between the MCP value and the first positive peak following the trigger pulse. SPP amplitude was taken as the average value over 200 ms centered on the main positive peak value (SPP) in the latency band between 350 and 650 ms from the left-hand trigger. This value was measured from the baseline. P200 and SPP latencies were both measured from the left-hand trigger pulse. PAN amplitude was taken as the average value over 200 ms centered around the main negative peak value (PAN) in the latency band between 350 and 850 ms from the left-hand trigger. This value was measured from the baseline. PAN latency was measured from the left-hand trigger pulse.

Analysis of the data reported in the various papers was based on multiple linear regression. Separate analyses of variance were done on the EMG and EEG data. The amplitudes of the tonic activity of left and right electromyograms, before the onset of movement and the rise time of the EMG, were regarded as the dependent variables for the statistical analysis of EMG data. The various macro potentials recorded from the diverse locations were the dependent variables for the statistical analysis of the MRBMs. The independent variables for both the EEG and the EMG data were age, subjects, and tests or time in the experiment. The analysis of variance was performed to test the significance of the multiple regression, yielding F values, and the significance of the individual partial regression coefficients, yielding t values.

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